



PROJECT REPORT

Breaking Barriers: Implementing Tech Solutions for Indigenous Communities in Northern Kenya

Indigenous Young Women-Led Collective | Northern Kenya ASAL

Implementation Period: September 2025

Locations: Turkana and Samburu Counties



1. Executive Summary

In September 2025, Beyond Borders implemented Breaking Barriers, a pilot project testing practical, low-cost technology solutions designed specifically for indigenous pastoralist communities in Northern Kenya. The project trained 22 young women to deploy, manage, and adapt three core tools: offline mapping applications for resource tracking, solar-powered community radio linkages, and simple digital market-information systems. Unlike standard tech-for-development models, Breaking Barriers required that every solution be maintainable without external technical support, function without reliable internet, and remain under community governance.

The pilot demonstrated that barriers to technology are not primarily about innovation — they are about design fit. When tools are adapted to low-connectivity, mobile, oral-culture contexts, and when indigenous women manage them, adoption rates increase and community trust is preserved.

22 Young women implementers	3 Tech solutions tested	6 Settlement hubs reached	100% Locally maintained
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2. Background and Rationale

Pastoralist communities in Northern Kenya face persistent barriers to information and communication: drought early warnings often arrive too late, women livestock traders lack market price data, and conflict mediation suffers from poor coordination across vast territories. Conventional tech solutions — smartphone apps, cloud-based platforms, biometric systems — assume levels of connectivity, literacy, and individual device ownership that do not exist in most ASAL settlements.

Beyond Borders designed Breaking Barriers to test whether appropriately simple, offline-first, collectively managed technologies could address these gaps without replicating patterns of dependency or extraction. The project focused on tools that could be repaired locally, powered by solar energy, and operated by women who understand both the technology and the social context.

3. Objectives

- Test three appropriate technology solutions in real pastoralist settlement conditions.
- Train a cohort of indigenous young women as technology implementers and troubleshooters.
- Assess whether community-managed tech hubs can sustain operation without permanent external technical support.
- Document which barriers (power, connectivity, literacy, gender norms) are most decisive for each tool type.
- Establish maintenance and governance protocols that keep technology under local control.

4. Approach and Methods

4.1 Tech solution selection

Through community consultation, three tools were selected:

- **Offline mapping (using Mapeo and paper-digital hybrid methods)** for tracking grazing, water points, and conflict-sensitive areas.
- **Solar-powered community radio linkages** connecting settlement hubs to existing FM stations for drought and market alerts.
- **Simple SMS-based market information** using basic feature phones to share livestock prices between women traders.

4.2 Training and deployment

Twenty-two young women underwent practical training in device management, solar charging systems, basic troubleshooting, and collective governance of shared equipment. Deployment occurred in six settlement hubs, with participants adapting workflows to local mobility patterns and household responsibilities.

4.3 Governance

Each hub established a technology management committee (majority women) responsible for device care, access scheduling, and reporting malfunctions. No data was uploaded to external servers without explicit collective consent.

5. Key Results and Outputs

Tangible outputs

- Six functional community tech hubs operating across Turkana and Samburu counties.
- 22 young women trained as technology implementers, capable of basic hardware troubleshooting and workflow adaptation.
- Three tested and documented tech solutions with ASAL-adapted user guides in Kiswahili and Turkana.
- Improved market information flow among women livestock traders in four settlement clusters.

Relational and capacity outcomes

- Demonstration that community-managed hubs can sustain operation for 8+ months without external technical visits.
- Increased confidence among women participants in negotiating with male-dominated livestock markets using price data.
- Clear evidence that offline-first tools outperform cloud-dependent alternatives in ASAL conditions.
- Community-developed maintenance checklists and governance norms now informing broader collective policy.

6. Challenges and Lessons Learned

- **Hardware durability in harsh environments.** Dust, heat, and rough transport damaged equipment faster than anticipated. Ruggedised or locally repairable devices are essential.
- **Solar dependency and weather.** Extended cloudy periods reduced charging capacity. Hybrid power strategies (solar + hand-crank or pedal options) warrant exploration.
- **Feature phone persistence.** Smartphones remain scarce; SMS and voice-based solutions reached far more people than app-based tools.
- **Gendered time poverty.** Women could not attend hub sessions during peak livestock or water-fetching periods. Scheduling must align with pastoralist labour calendars.

— **External assumptions die hard.** Some partner organisations continued pushing app-based solutions that failed in testing. Community implementers needed advocacy skills to resist inappropriate donor preferences.

Key lesson: *The barrier is rarely the community. The barrier is technology designed for radically different contexts. Breaking barriers means redesigning the tool, not demanding the environment change.*

7. Contribution to Beyond Borders' Work

Breaking Barriers proved that indigenous young women can implement and sustain technology in some of Kenya's most infrastructure-poor regions. It generated practical toolkits, maintenance protocols, and evidence that now anchor the collective's partnerships and advocacy. The project also confirmed that Beyond Borders' offline-first, community-governed approach is not a limitation but a design strength.

8. Looking Ahead

Beyond Borders will expand the hub network, deepen repair and maintenance training, and explore mesh networking for inter-settlement communication. All future tool adoption will be filtered through the Breaking Barriers criteria: offline-capable, locally maintainable, collectively governed, and gender-responsive.